

HILO 5 S, HI, USA

WMO: 912960

Lat: 19.6450N

Lon: 155.0827W

Elev: 622

StdP: 14.37

Time Zone: -10.00 (W10)

Period: 05-19

WBAN: 21515

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	58.0	59.3	53.6	62.5	62.8	55.2	66.4	63.6	6.9	77.3	6.6	76.9	0.2	N/A	0.187

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	82.3	72.6	81.2	72.0	80.2	71.5	75.3	79.5	74.4	78.5	73.6	77.6	4.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	129.8	77.6	73.2	126.2	76.8	72.4	122.7	76.1	39.2	79.7	38.4	78.4	37.6	77.9	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years					
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.5	5.7	4.8	DB	55.1	84.5	0.9	1.9	54.5	85.9	54.0	87.0	53.5	88.0	52.8	89.4
			WB	52.8	76.8	1.0	1.6	52.1	78.0	51.5	78.9	51.0	79.8	50.3	80.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.4	68.0	67.9	68.3	69.2	70.5	71.4	72.4	73.0	72.8	72.1	70.5	69.0
	DBStd	2.68	1.93	2.54	2.28	2.03	1.83	1.48	1.82	1.84	1.93	1.86	2.00	1.97
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	9	2	4	3	0	0	0	0	0	0	0	0	0
	CDD50	7457	558	501	566	577	634	641	693	714	683	686	615	589
	CDD65	1993	95	85	104	128	169	191	228	249	233	221	165	125
	CDH74	5905	206	217	220	249	450	535	824	909	962	728	394	211
	CDH80	246	4	10	7	5	12	12	31	51	67	33	10	4
(14) Wind	WSAvg	1.4	1.4	1.5	1.5	1.6	1.6	1.7	1.6	1.4	1.3	1.1	1.2	1.3
(15) Precipitation	PrecAvg	145.0	10.9	13.0	15.5	12.2	9.5	8.4	11.0	10.9	10.2	11.7	16.7	15.1
	PrecMax	237.5	32.4	36.5	41.5	22.8	26.5	27.1	23.5	23.0	24.7	25.5	45.8	29.5
	PrecMin	76.6	0.4	1.4	1.4	5.0	2.4	3.1	3.1	4.4	1.9	3.3	1.4	4.4
	PrecStd	29.4	7.8	8.3	9.8	4.2	5.1	4.5	4.6	4.4	5.7	5.0	10.1	7.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.4	81.5	80.8	80.6	81.5	81.7	82.4	83.5	83.9	82.9	81.5	80.3
		MCWB	68.2	68.6	69.8	69.9	70.5	71.2	73.2	74.5	74.4	73.2	72.5	69.9
	2%	DB	78.2	78.8	78.5	78.3	79.6	79.7	80.8	81.4	81.9	80.8	79.5	77.9
		MCWB	67.8	68.9	68.9	68.8	69.8	70.4	72.1	73.1	73.1	72.5	71.3	69.4
	5%	DB	76.4	76.8	76.7	76.9	78.1	78.3	79.7	80.0	80.4	79.5	77.9	76.4
		MCWB	67.5	68.1	68.4	68.4	69.3	69.9	71.5	72.3	72.4	72.0	70.4	68.9
	10%	DB	74.7	74.7	74.6	75.4	76.8	77.2	78.4	78.6	79.1	78.1	76.5	75.0
		MCWB	66.9	67.2	67.6	68.1	68.9	69.5	71.1	71.8	71.8	71.4	69.9	68.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.0	72.0	71.8	72.3	72.5	73.2	75.5	76.4	76.7	75.6	73.5	72.6
		MCDB	76.5	77.2	77.8	77.6	77.5	78.2	79.9	80.9	82.1	79.3	78.3	76.9
	2%	WB	69.6	70.4	70.5	70.8	71.5	72.1	74.2	75.2	75.2	74.3	72.6	71.1
		MCDB	75.0	75.8	76.1	75.2	76.5	77.0	78.2	79.1	79.6	78.3	77.1	75.0
	5%	WB	68.9	69.4	69.4	69.9	70.7	71.3	73.2	74.3	74.3	73.3	71.7	70.1
		MCDB	74.0	74.9	74.4	74.4	75.6	76.1	77.1	77.9	78.6	77.5	75.8	73.9
	10%	WB	68.1	68.6	68.6	69.2	69.9	70.6	72.3	73.3	73.3	72.6	71.0	69.2
		MCDB	73.1	73.6	73.3	73.7	74.7	75.3	76.5	76.8	77.5	76.7	74.9	72.8
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	13.4	12.5	12.1	12.8	12.5	12.4	12.0	13.3	12.6	12.3	12.3
		MCDBR	16.7	16.9	16.1	14.5	14.8	14.1	14.0	13.8	14.8	14.3	14.3	14.6
		MCWBR	9.2	8.8	8.4	6.9	7.3	6.8	6.7	6.9	7.7	7.9	8.1	8.2
	5% WB	MCDBR	14.4	13.9	14.4	12.3	12.7	12.1	11.9	10.9	12.6	12.5	12.3	11.5
		MCWBR	8.8	8.0	8.2	6.7	7.2	6.6	6.9	6.7	7.5	7.5	7.8	7.5
(40) Clear-Sky Solar Irradiance	taub		0.359	0.361	0.378	0.416	0.397	0.385	0.391	0.377	0.375	0.373	0.369	0.354
	taud		2.575	2.568	2.517	2.373	2.463	2.488	2.482	2.534	2.538	2.569	2.582	2.613
	Ebn at Noon		292	298	295	283	285	285	284	290	292	290	287	291
	Edh at Noon		30	31	34	40	36	35	35	34	33	31	29	28
(44) All-Sky Solar Radiation	RadAvg		1423	1544	1690	1860	1953	2007	2043	1985	1849	1617	1391	1290
	RadStd		86	103	183	127	122	72	91	95	76	65	79	122

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+225	+225

Nomenclature: See separate page